

# An AI-Driven Adaptive Cybersecurity Awareness and Phishing Simulation Platform

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## Abstract

- A safe, classroom-focused cybersecurity awareness platform that combines adaptive phishing simulations, assessments, progress tracking, and retrieval-grounded security knowledge. The system uses React/Vite, FastAPI, Node.js, Spring Boot, PostgreSQL + pgvector, MongoDB and Redis.

The adaptive engine uses topic accuracy, recency and response time to select the next practice scenario and bounded difficulty.

## Problem Statement & Objectives

- Problem: Conventional awareness training can be static, repetitive, and difficult to personalize while live phishing activity introduces safety risks.
- Objective 1: Provide safe fictional phishing simulations and assessments.  
Objective 2: Adapt topic and difficulty using learner performance signals.  
Objective 3: Track progress, badges, analytics and knowledge retrieval.  
Scope: Student learning, educator views, simulations, assessments, analytics and secure service APIs.

## Existing vs Proposed System

| Feature         | Existing System                 | Proposed System                            |
|-----------------|---------------------------------|--------------------------------------------|
| Training        | Static awareness content        | Adaptive topic & difficulty practice       |
| Simulation      | Live or risky phishing activity | Fictional, contained scenarios             |
| Personalization | Same path for learners          | Performance-based next scenario            |
| Knowledge       | Generic lookup                  | pgvector retrieval + grounded explanations |
| Monitoring      | Limited visibility              | Progress, badges, analytics & audit trail  |

## Feasibility Study

- Technical: Uses React/Vite, FastAPI, Node.js, Spring Boot, PostgreSQL/pgvector, MongoDB, Redis and Docker Compose.
- Economic: Uses open-source technologies and local/containerized deployment for classroom demonstration.  
Operational: Fictional scenarios, disabled external navigation and no credential collection reduce operational risk.  
Schedule: Modular services and documented deployment scripts support incremental development and testing.

## Results & Testing

- Core UI includes registration/login and adaptive learning flows.
- Database design includes relational learning state, attempts, badges, campaigns and knowledge documents.  
Project defines FastAPI, Node.js and Spring tests plus a frontend production build.  
Safe simulation controls prevent external navigation and credential collection.

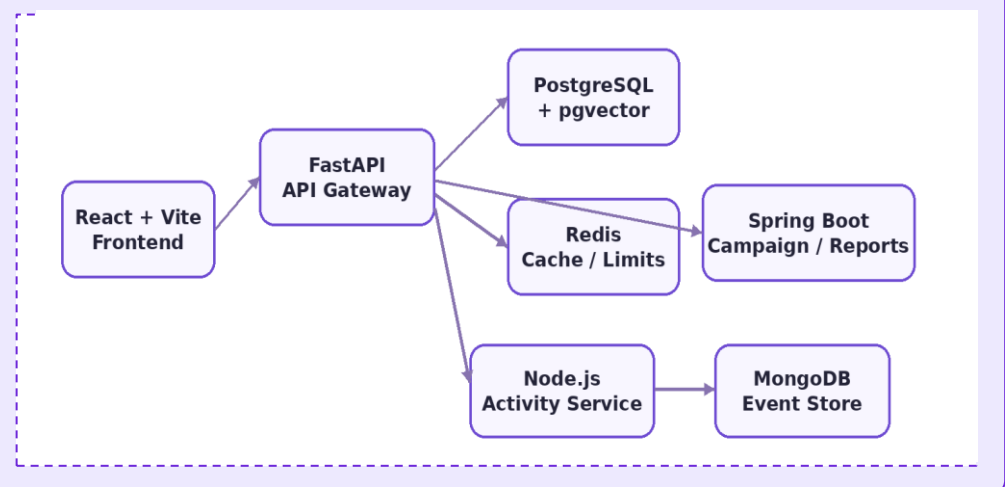
## Conclusion

- The project delivers a modular cybersecurity awareness platform that connects adaptive learning, safe phishing simulations, assessments, knowledge retrieval and analytics through secure APIs and persistent data services.

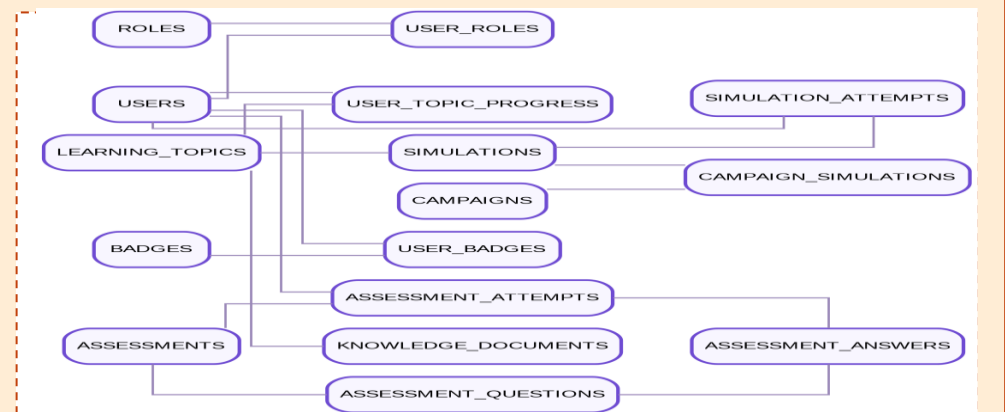
## Tools & Technologies

- Front-End: React 18, Vite, React Router, Axios, Recharts
- API / Back-End: FastAPI, Node.js + Express, Spring Boot 3.3.5, REST APIs  
Data: PostgreSQL 16 + pgvector, MongoDB 7, Redis 7  
Security: bcrypt password hashing, signed JWT, role checks, throttling  
DevOps: Docker Compose, Prometheus, Grafana, OpenTelemetry, VS Code

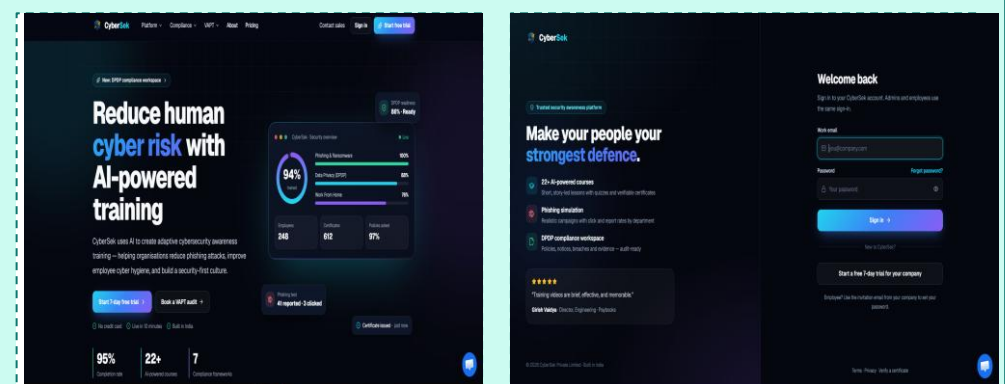
## System Architecture



## ER Diagram & Database Design



## Implementation Screenshots



## Future Scope

- Cohort-level learning analytics and outcome comparison.
- More adaptive signals and tunable recommendation models.
- Expanded educator dashboards and reporting.
- Production deployment, scaling and richer observability.